

METHODOLOGY

Data Collection

- Soil moisture data from capacitive sensor.
- Temperature and humidity data from DHT11.

Decision Making

- Compare moisture level with predefined threshold.

Execution

- Activate relay when soil is dry.
- Start pump until moisture reaches desired level.

Monitoring Cycle

- Sensor reading every 5 minutes.

SOFTWARE IMPLEMENTATION

Development Environment

- Arduino IDE
- Programming Language: C++

Software Functions

- Read sensor values
- Process environmental data
- Compare moisture threshold
- Control relay and pump
- Display readings through Serial Monitor

Sampling Interval

Every 5 minutes

AI MODEL DEVELOPMENT

Current Approach

This project does not use Artificial Intelligence or Machine Learning.

Control Strategy

- Sensor-based automation.
- Rule-based decision making.
- Fixed moisture threshold.
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Moisture Calibration

Experimental testing determines the optimal watering threshold for Pothos plants.

Future Improvement

- AI prediction of watering schedules.
- Plant growth analysis.
- Cloud-based monitoring.